

07.04 Multiplying and Dividing Rational Expressions

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Rational Expressions And Equations	State denominator restrictions before simplifying or solving.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Multiplying and Dividing Rational Expressions

1.

$$\frac{x^2 - 9}{x^2 + 3x}$$

2.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

3.

$$\frac{x + 1}{x^2 - 4}$$

4.

$$\frac{x^2 - 9}{x^2 + 3x}$$

5.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

6.

$$\frac{x+1}{x^2-4}$$

7.

$$\frac{x^2-9}{x^2+3x}$$

8.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

9.

$$\frac{x+1}{x^2-4}$$

10.

$$\frac{x^2-9}{x^2+3x}$$

11.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

12.

$$\frac{x+1}{x^2-4}$$

13.

$$\frac{x^2-9}{x^2+3x}$$

14.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

15.

$$\frac{x+1}{x^2-4}$$

16.

$$\frac{x^2-9}{x^2+3x}$$

17.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

18.

$$\frac{x+1}{x^2-4}$$

19.

$$\frac{x^2-9}{x^2+3x}$$

20.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

B. Apply the key formulas

1.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

2.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

3.

LCD clears denominators

4.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

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$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

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